c. **Smears:** Here the specimen is in the fluid (blood, microbial cultures etc.,) are scraped, brushed or aspirated from surface of organ. Example: Epithelial cells.

d. **Sections:** Free hand sections from a specimen and thin sections are selected, stained and mounted on a slide. Example: Leaf and stem of plants.

6.9.2 Recording the Observations

The observations made through a microscope can be recorded by hand diagrams or through microphotographs.

Hand diagrams: Hand diagrams are drawn using ordinary pencil by observing the slide and drawing manually.

Microphotograph: Images of structures observed through microscopes can be further magnified, projected and saved by attaching a camera to the microscope by a microscope coupler or eyepiece adaptor. Picture taken using a inbuilt camera in a microscope is called **microphotography** or **microphotograph**.

6.9.3 Staining Techniques

Staining is very important to observe different components of the cell. Each component of the cell has different affinity towards different stains. The technique of staining the cells and tissue is called '**histochemical staining**' or '**histochemistry**'.

Common stains used in Histochemistry

S. No.	Stain	Colour of staining	Affinity
1.	Eosin	Pink, Red	Cytoplasm, cellulose
2.	Acetocarmine/ Haematoxylin	Pink/ Red	Nucleus, Chromosomes
3.	Methylene Blue	Blue	Nucleus
4.	Saffranine	Red	Cell wall (Lignin)
5.	Cotton blue	Blue	Fungal Hyphae
6.	Sudan IV, Sudan Black	Scarlet Red/Black	Lipids
7.	Coomasie brilliant Blue	Blue	Protein
8.	Janus Green	Greenish Blue	Mitochondria
9.	I_2KI	Bluish black to brown	Starch
10.	Toluidine blue	Blue, greenish blue	Xylem, Parenchyma & Epidermis

Summary

Cell is the fundamental unit of all organisms which was identified 300 years ago. Microscope offers scope for observing smaller objects and organisms. It works on the principle of light and lenses. Different microscope offers clarity in observing objects depending on the features to be observed. Micrometric techniques are used in measurement of microscopic objects. Electron microscopes are used in understanding the ultra-structural details of cell. Cell theory and doctrine states that all organism are made up of cell and it contains genetic material. Protoplasm theory explains nature and different properties of protoplasm. Cell size and shape differ from type of tissue or organs and organisms. Based on cellular organization and nuclear characters the